Correlated Thompson Sampling via LLM-Derived Covariance Structure for Combinatorial Semi-Bandits

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Abstract

We study combinatorial semi-bandits augmented with a large language model (LLM) oracle. Existing methods inject LLM output directly into the posterior as priors, pseudo-observations, or arm recommendations. We show that this belief-injection approach is fragile because LLM accuracy depends on the estimates it receives as input: in one configuration, the same LLM achieves 4/5 top-m accuracy when warm-started with Thompson sampling estimates and 0/5 with round-robin estimates. We propose using the LLM for *correlation structure* instead. A single cluster query is converted to a positive-definite covariance matrix via an RBF kernel over cluster ranks; only the sampling step changes, and the Beta posteriors are updated from real rewards alone. On synthetic Bernoulli instances (T=2,500, n=320), CORRCTS-FULL reduces regret by 18.9% versus CTS ($p < 0.0001$, Holm-Bonferroni), of which 12.7 points are attributable to LLM cluster quality via a random-cluster ablation. On a synthetic categorical recommendation benchmark with planted latent category structure (d=200, m=5), the reduction reaches 33.8% with a 135/0 win/loss ratio, while warm-start CTS offers no benefit. This benchmark tests whether the kernel exploits latent cluster structure when present; it does not test whether the LLM discovers structure in real click logs.

1. Introduction

We study the stochastic combinatorial semi-bandit problem. A learner selects m arms from a ground set of d arms each round, observes a Bernoulli reward for each selected arm, and minimizes cumulative regret over T rounds. Combinatorial Thompson Sampling (CTS) (Wang & Chen, 2018) maintains independent Beta posteriors and samples each arm's mean independently. This ignores any structure among arms, even when structure is readily available.

A natural source of structural information is a large language model. Existing LLM-augmented bandit methods inject the LLM's output as a *belief*: Krishnamurthy et al. (2024) study in-context exploration; Alamdari et al. (2024); Bayley et al. (2025) warm-start bandits with LLM-generated priors; Sun et al. (2025) mix LLM arm recommendations with Thompson sampling. In each case, the LLM's output enters the posterior as a prior shift, a pseudo-observation, or a direct arm recommendation. A separate line, Cao et al. (2026), uses LLM guidance in bandit recourse for personalized treatment planning and is orthogonal to our setting.

Belief injection is fragile, and we trace the fragility to a specific phenomenon that existing models do not capture: the LLM's answer quality depends on the estimates it receives as input. In a representative configuration ($d = 25$, $m = 5$, 20 seeds), GPT-5.4 achieves 4 out of 5 top-m accuracy after 30 rounds of CTS warmup, which concentrates pulls on promising arms, and 0 out of 5 after 30 rounds of round-robin, which spreads pulls uniformly. Same model, same prompt template, same horizon; only the input estimates differ. This single-configuration diagnostic is illustrative, not conclusive; replication across (d, m) settings and warmup strategies is forthcoming in the camera-ready. Oracle quality is therefore endogenous to the bandit's learning trajectory. When $\hat{\mu}$ is noisy, belief injection amplifies errors; when $\hat{\mu}$ is accurate, the LLM's information is redundant. The window in which belief injection helps is narrow, and existing theoretical models for LLM-bandits that treat oracle quality as an exogenous parameter ϵ (Sun et al., 2025) do not reach inside this window.

We propose a different design: use the LLM for *structure*, not *belief*. We ask the LLM which arms behave similarly and obtain a partition into clusters. When we sample posterior values, we draw *correlated* samples for same-cluster arms rather than independent ones. Observing arm 3's reward now implicitly informs our sampling of arms 7 and 12 whenever the LLM has placed them in the same cluster. The correlation matrix is an RBF kernel over a cluster-rank em-

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bedding, which is positive definite by construction (Proposition H.2). The Beta posteriors are never touched by the LLM; only the sampling step changes. When the LLM is wrong, bad off-diagonal entries add noise to the samples; the diagonal and the posterior means remain uncontaminated, and performance degrades gracefully to CTS.

Our contributions are the following. We design CORRCTS-FULL, a correlated Thompson sampling algorithm built from a single LLM cluster query. On 16 synthetic configurations (n=320 paired trials, T=2,500), CORRCTS-FULL reduces regret by 18.9% versus CTS ($p < 0.0001$, Holm-Bonferroni corrected). A random-cluster ablation isolates 12.7 percentage points of this gain to LLM cluster quality, and a block-diagonal variant does not beat random clusters, showing that the inter-cluster structure captured by the RBF kernel drives the improvement. We develop two data-adaptive variants, CORRCTS-PRUNE and CORRCTS-KM, that refresh the covariance from observed data and sustain a 6.1–6.8% advantage at T=25,000 (n=480), validated on held-out configurations. On a synthetic categorical recommendation benchmark with planted category structure (MIND-style, d=200, m=5), the reduction reaches 33.8% with a 135/0 win/loss ratio against CTS; warm-start CTS, the belief-injection baseline, provides no benefit on this task.

2. Related Work

Combinatorial semi-bandits. Chen et al. (2013) introduced the CMAB framework with CUCB, achieving $O(d \log T/\Delta_{\min})$ regret. Kveton et al. (2015) proved tight $O(d \log T/\Delta_{\min})$ bounds for the semi-bandit setting. Combes et al. (2015) proposed ESCB, the canonical structured-combinatorial baseline: given a linear feature matrix $X \in \mathbb{R}^{d\times p}$ with $\mu = X\theta$, ESCB maintains a confidence ellipsoid on θ and selects actions by a concentration index, achieving an $\sqrt{m}$ improvement in the dependence on m. Our LLM oracle returns a partition rather than a feature matrix that ESCB consumes directly; a one-hot cluster-indicator extension is feasible but imposes a rigid within-cluster homogeneity assumption that our RBF construction explicitly relaxes, so we treat ESCB as out of scope for the head-to-head comparison and discuss the extension in Appendix L. Wang & Chen (2018) analyzed CTS and proved $O(d \log T/\Delta_{\min})$ Bayesian regret; Perrault et al. (2020) refined these bounds. We build on CTS as the base algorithm.

LLM-augmented bandits. Krishnamurthy et al. (2024) showed that LLMs struggle with in-context exploration for stochastic bandits but can provide prior information. Alamdari et al. (2024) and Bayley et al. (2025) warm-start bandits with LLM-generated priors, treating LLM outputs as pseudo-observations. Sun et al. (2025) designed TS-LLM, which mixes LLM arm recommendations with Thompson sampling under a decaying probability; we include TS-LLM as a baseline and find it ranks second overall in Experiment 1. Cao et al. (2026) integrate LLM guidance with bandit recourse for personalized treatment planning; their setting differs from ours, and we do not benchmark against them directly. Liu et al. (2024) use LLMs for Bayesian optimization. Our approach differs in what it extracts and where it injects it: we extract *structure* (a partition) rather than beliefs (arm rankings or pseudo-rewards), and we place it in the sampling covariance rather than the posterior parameters.

Latent structure and correlated arms (non-LLM). Hong et al. (2020) study latent bandits where arm parameters share a low-dimensional structure learned online; we differ in that structure is supplied by an external oracle rather than inferred. Gupta et al. (2021) derive regret bounds when inter-arm correlations are known a priori; our contribution lies in reading that structure off a single LLM partition and validating it empirically. Swersky et al. (2013) use kernels seeded by task metadata in multi-task BO; we adopt the same kernel-on-discrete-features idea but drive it from the LLM.

Correlated sampling and GP bandits. When m arms are selected from d, independent sampling requires all m optimal arms to receive high posterior draws simultaneously, whose probability decreases rapidly with m; positively correlated samples over correct clusters let a single favorable draw lift all cluster-mates together. GP-TS (Srinivas et al., 2010) uses kernel covariance with a regret bound in the information gain γ_T; we apply the kernel idea to discrete combinatorial actions with LLM-derived features and an approximated Beta posterior. Mixed-effect Thompson sampling (Aouali et al., 2023) requires a known feature matrix; we let the LLM supply it. Graph-based online clustering (Gentile et al., 2014) maintains edges among users with overlapping confidence intervals; our CORRCTS-PRUNE variant borrows this idea for kernel refresh. CLUB and related no-LLM online-clustering baselines are comparators we defer to the camera-ready.

Learning-augmented algorithms. The algorithms-with-predictions framework (Lykouris & Vassilvitskii, 2018) studies how to use untrusted machine-learned advice while retaining worst-case guarantees. Our credibility-gated ROBUSTCORRCTS follows this philosophy: it interpolates between LLM-guided correlated sampling and standard CTS based on an empirical validation signal, and Proposition 4.4 bounds its worst-case excess regret over CTS-Gaussian by $O(T/\Delta_{\text{check}})$.

3. Problem Setup

Combinatorial semi-bandit. There are d arms, each with unknown Bernoulli mean $\mu_i \in [0,1]$. At each round $t = 1, \ldots, T$, the learner selects $S_t \subseteq [d]$ with $|S_t| = m$. For each arm $i \in S_t$, a reward $X_{i,t} \sim \text{Bernoulli}(\mu_i)$ is observed independently. The optimal action is $S^\star = \arg\max_{S:|S|=m} \sum_{i \in S} \mu_i$, and the cumulative regret is

$$R(T) = \sum_{t=1}^{T} \Big(\sum_{i \in S^\star} \mu_i - \sum_{i \in S_t} \mu_i \Big). \tag{1}$$

LLM oracle. At a warmup time T_w, the learner may query an LLM oracle $\mathcal{O}$. The oracle receives the current empirical means $\hat{\mu} = (\hat{\mu}_1, \ldots, \hat{\mu}_d)$ and returns a partition $\mathcal{C} = \{C_1, \ldots, C_K\}$ of $[d]$ into K clusters, where same-cluster arms are deemed similar by the LLM. The oracle's quality is unknown to the learner and may depend on $\hat{\mu}$. Each query incurs a token cost; a single cluster query costs roughly 600 tokens ($0.003 on GPT-5.4).

Combinatorial Thompson Sampling. CTS (Wang & Chen, 2018) maintains independent Beta posteriors $\text{Beta}(\alpha_i, \beta_i)$ for each arm, initialized at $\text{Beta}(1,1)$. At each round it draws $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ independently and plays $S_t = \arg\max_{S:|S|=m} \sum_{i \in S} \theta_i$. After observing rewards, $\alpha_i \leftarrow \alpha_i + X_{i,t}$ and $\beta_i \leftarrow \beta_i + (1 - X_{i,t})$ for $i \in S_t$. Our algorithms replace the independence of $\theta_1, \ldots, \theta_d$ with an LLM-informed correlation structure; the posterior update is unchanged.

4. Algorithms

4.1. CorrCTS: the framework

The CorrCTS framework has three steps, executed once at round T_w. *Query:* send $\hat{\mu}$ to the LLM and receive clusters $\mathcal{C} = \{C_1, \ldots, C_K\}$. *Build covariance:* convert $\mathcal{C}$ to a $d \times d$ positive-definite correlation matrix Σ via an RBF kernel over cluster ranks. *Sample:* at each subsequent round, compute the Beta posterior mean $\bar{\mu}_i = \alpha_i/(\alpha_i + \beta_i)$ and standard deviation $\sigma_i = \sqrt{\alpha_i \beta_i / ((\alpha_i + \beta_i)^2 (\alpha_i + \beta_i + 1))}$, draw $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$, and set $\theta_i = \bar{\mu}_i + \sigma_i \cdot (L\boldsymbol{\epsilon})_i$ where L is the Cholesky factor of Σ. Play $S_t = \arg\max_{S:|S|=m} \sum_{i \in S} \theta_i$.

The sampling distribution is a Gaussian approximation to the Beta posterior, moment-matched to $(\bar{\mu}_i, \sigma_i^2)$; Proposition H.1 bounds the approximation error by $O(1/\sqrt{n_i})$ in total variation. When $\Sigma = I_d$, the algorithm reduces to CTS-Gaussian (Proposition 4.2). When Σ has off-diagonal entries, same-cluster arms receive correlated noise: a high sample for one arm raises the samples of its cluster-mates.

A rough calculation shows why this matters. Consider m optimal arms in the same cluster, each with per-round probability $q \in (0, 1/2)$ of a sample above threshold under independent CTS. The probability that all m samples exceed the threshold simultaneously is q^m under independence. Under a correlated sample with rank-one correlation $\rho \in (0,1)$, the same event has probability at least $q \cdot (1 - O(1-\rho))$ once ρ is close to 1, because the common noise component dominates. The exponential-in-m factor disappears in the correlated regime. This heuristic matches the empirical finding that the gain over CTS grows with m and with problem size in the per-(d, m) breakdown (Appendix E).

Algorithm 1 CorrCTS-Full

Require: arms d, subset size m, warmup T_w, kernel params $\rho_{\max}, \ell$
$\alpha_i \leftarrow 1, \beta_i \leftarrow 1$ for all i
for $t = 1, \ldots, T$ **do**
  if $t \le T_w$ **then**
    $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ independently
  else if $t = T_w + 1$ **then**
    $\mathcal{C} \leftarrow$ LLM_ClusterQuery$(\hat{\mu})$
    $f_i \leftarrow$ ClusterRank$(i, \mathcal{C})$
    $\Sigma_{ij} \leftarrow \rho_{\max} \exp(-(f_i - f_j)^2/(2\ell^2))$, $\Sigma_{ii} \leftarrow 1$
    $L \leftarrow$ Cholesky$(\Sigma + 10^{-3} I_d)$
  end if
  if $t > T_w$ **then**
    $\bar{\mu}_i \leftarrow \alpha_i/(\alpha_i + \beta_i)$, $\sigma_i \leftarrow \sqrt{\alpha_i \beta_i / ((\alpha_i + \beta_i)^2 (\alpha_i + \beta_i + 1))}$
    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$; $\theta_i \leftarrow \bar{\mu}_i + \sigma_i (L\boldsymbol{\epsilon})_i$
  end if
  Play $S_t = \text{top-}m(\boldsymbol{\theta})$; observe $X_{i,t}$ for $i \in S_t$
  $\alpha_i \leftarrow \alpha_i + X_{i,t}$, $\beta_i \leftarrow \beta_i + (1 - X_{i,t})$ for $i \in S_t$
end for

4.2. CorrCTS-Full

CorrCTS-Full uses a full (not block-diagonal) $d \times d$ RBF kernel covariance. Order the clusters from 0 to $K-1$ by the LLM's output order; assign each cluster rank r_k and normalize $\tilde{r}_k = r_k/(K-1) \in [0,1]$. Each arm $i \in C_k$ receives feature $f_i = \tilde{r}_k$. The correlation matrix is the RBF kernel

$$\Sigma_{ij} = \rho_{\max} \cdot \exp\Big(-\frac{(f_i - f_j)^2}{2\ell^2} \Big), \quad i \neq j, \tag{2}$$

with $\Sigma_{ii} = 1$, $\rho_{\max} \in (0,1)$, and lengthscale $\ell > 0$. Same-cluster arms have $f_i = f_j$ and correlation $\rho_{\max}$. Adjacent clusters get smaller positive correlation; distant clusters get near zero. We add jitter $10^{-3} I_d$ before Cholesky for numerical stability; positive definiteness is guaranteed by Proposition H.2.

A simpler alternative would set Σ block-diagonal: a fixed correlation ρ within each cluster, zero across clusters. This discards the ordering of clusters. CorrCTS-Full's RBF kernel preserves this ordering: clusters that the LLM placed

adjacently (similar estimated performance) retain positive correlation. The ablation in Section 5.1 shows block-diagonal correlation alone does not beat random clusters; the inter-cluster smoothing is what makes the LLM's output useful.

4.3. Data-adaptive variants for long horizons

CORRCTS-FULL's kernel is built at $T_w = 30$ and never refreshed. At $T = 2{,}500$ this is adequate; at $T = 25{,}000$ the static kernel decays as the learner accumulates more precise estimates than the ones the LLM saw. The failure mode is concrete: arms whose true means differ substantially may have been placed in the same LLM cluster because their $\hat{\mu}$ at $T_w = 30$ looked similar, and the kernel then binds their samples together long after the learner has enough observations to tell them apart. We address this with two variants that keep a single LLM query and periodically refine the kernel from observed data.

CORRCTS-PRUNE progressively prunes edges. Every 500 rounds, for each pair (i, j), if $|\hat{\mu}_i - \hat{\mu}_j|$ exceeds a concentration-based threshold $\beta\sqrt{1/n_i + 1/n_j}$, the entry Σ_{ij} is shrunk by a factor of 10. Arms whose means are well-separated lose their LLM-imposed correlation; arms that remain statistically indistinguishable retain it. This is related to the graph-based clustering of CLUB (Gentile et al., 2014).

CORRCTS-KM blends the LLM kernel with a data-driven kernel. At $t \in \{2{,}000,\ 10{,}000\}$ it runs $\sqrt{n_i}$-weighted k-means on $\hat{\mu}$ to obtain a data partition, converts it to an RBF kernel with the same construction, and blends: $\Sigma_t = w(t)\Sigma_{\text{LLM}} + (1 - w(t))\Sigma_{\text{data}}$ with $w(t) = \max(0.2,\ 1 - t/12{,}000)$. No additional LLM queries are used. Both variants add negligible overhead (one eigendecomposition per update).

4.4. Safety properties

Before adding the credibility gate, we collect two safety properties that hold for CORRCTS-FULL regardless of the LLM's output. The proofs are in Appendix H.

Proposition 4.1 (Posterior integrity)**.** *The Beta posteriors $(\alpha_i, \beta_i)_{i\in[d]}$ in* CORRCTS-FULL *are updated identically to standard CTS. The LLM covariance Σ enters the sampling distribution only; it never modifies (α_i, β_i).*

This is the structural safety net. An adversarial LLM can corrupt Σ arbitrarily and the posterior stays correct: bad off-diagonals in Σ add correlated noise to the samples, but the marginal distribution of θ_i remains $\mathcal{N}(\bar{\mu}_i, \sigma_i^2)$. Belief-injection algorithms do not have this property; a single bad recommendation from the LLM shifts the posterior and requires additional real observations to correct.

Proposition 4.2 (CTS recovery and approximation error)**.** *When $\Sigma = I_d$,* CORRCTS-FULL *reduces to CTS with Gaussian-approximated posteriors: $\theta_i \sim \mathcal{N}(\bar{\mu}_i, \sigma_i^2)$ independently across arms. Under this reduction, $d_{TV}(\text{Beta}(\alpha_i, \beta_i), \mathcal{N}(\bar{\mu}_i, \sigma_i^2)) \le C/\sqrt{n_i}$ for a constant $C = C(\alpha_0, \beta_0)$ depending only on lower bounds $\alpha_i \ge \alpha_0, \beta_i \ge \beta_0$; because Algorithm 1 initializes $\alpha_i = \beta_i = 1$ and only increments, C is universal across the algorithm's trajectory.*

The Gaussian approximation introduces $O(1/\sqrt{n_i})$ total-variation error per arm, which vanishes once each arm has been pulled enough times. In practice the approximation is faithful within 10 rounds of first pulling an arm (the Berry–Esseen rate); the experiments in Section 5.1 run for at least 2,500 rounds, so this term is negligible relative to the LLM-driven gains.

4.5. Bayesian regret

Under the Gaussian approximation of Proposition 4.2, we can reduce CORRCTS-FULL to combinatorial Gaussian-process Thompson sampling and import a known Bayesian regret bound.

Theorem 4.3 (Bayesian regret of CORRCTS-FULL, informal)**.** *Fix $n_0 \ge 2$ and let $T_0 = \min\{t : \forall i, n_i^{(t)} \ge n_0\}$. Assume the Gaussian-approximation regime of Proposition H.1 holds for $n_i \ge n_0$ and the LLM returns a partition with K clusters. Then the Bayesian cumulative regret of* CORRCTS-FULL *satisfies*

$$\mathbb{E}[R(T)] \;=\; m \cdot \mathbb{E}[T_0] \;+\; \tilde{O}\Big(m\sqrt{K\,T\log T}\Big)\,,$$

where the first term absorbs the early rounds before the reduction to Gaussian combinatorial GP-TS is invoked. The asymptotic rate improves over independent CTS's $\tilde{O}(m\sqrt{d\,T\log T})$ by a factor $\sqrt{d/K}$ whenever the LLM partition captures the latent structure.

The proof has four ingredients (details in Appendix I): an early-rounds term $m \cdot \mathbb{E}[T_0]$ (which uses Algorithm 1's initialization $\alpha_i = \beta_i = 1$ to pin the Berry–Esseen constant in Proposition H.1); a coupling between CORRCTS-FULL and Gaussian combinatorial TS with the same Σ; an explicit information-gain bound $\gamma_T = O(K \log T)$ derived from the eigendecomposition $\Sigma = \rho_{\max} K_{\text{RBF}} + (1 - \rho_{\max}) I_d$ (rank-K Gram plus ridge, via the finite-rank argument of Srinivas et al., 2010, Lemma 5.4); and invocation of Sandberg et al. (2025)'s combinatorial GP-TS bound $\tilde{O}(m\sqrt{\gamma_T T \log T})$. The same kernel-on-discrete-features idea appears in multi-task BO (Swersky et al., 2013); we differ in that the kernel is read off a single LLM partition. The rate interpolates correctly: $K = d$ recovers the CTS rate, and smaller K scales the effective dimension from d to the number of cluster identities the LLM exposes.

A scoped lower bound. On two-cluster Bernoulli instances with known cluster identity and gap $\Delta_{\min}$, any sampler whose per-round distribution factorizes as a product of arm-conditional distributions (CTS, warm-start CTS, CUCB, and any algorithm playing top-m on independent posterior draws) incurs expected regret at least $\Omega(d \log T/\Delta_{\min})$. The argument is a per-arm Lai–Robbins bound (Lai & Robbins, 1985): each of the $d - d/K$ suboptimal arms must be pulled $\Omega(\log T/\Delta^2_{\min})$ times to be distinguished from the optimum using its own history, and an arm-factorized sampler has no cross-arm information channel. CORRCTS-FULL with cluster-separating Σ couples arms inside a cluster and achieves $O((d/K)\log T/\Delta_{\min})$ (Appendix J). The bound separates CORRCTS-FULL from arm-factorized samplers. It does not separate CORRCTS-FULL from algorithms that mix LLM recommendations directly into the action distribution (TS-LLM) or maintain non-factorized posteriors. A lower bound matching the $\tilde{O}(m\sqrt{KT\log T})$ upper bound of Theorem 4.3 remains open.

4.6. RobustCorrCTS

ROBUSTCORRCTS adds a credibility gate. If the LLM returns poor clusters, correlated sampling could hurt by inducing spurious dependencies. ROBUSTCORRCTS maintains a credibility weight $w_t \in [0, 1]$ initialized at 1. Every Δ_{check} rounds it computes an overlap score between the LLM's highest-ranked cluster and the empirically best m arms, and updates $w_t \leftarrow 0.7w_{t-1} + 0.3\,\text{overlap}_t$. Each round draws two independent samples, a correlated $\boldsymbol{\theta}^{\text{corr}}$ and an independent $\boldsymbol{\theta}^{\text{CTS}}$, and plays the top-$m$ of $w_t\boldsymbol{\theta}^{\text{corr}} + (1 - w_t)\boldsymbol{\theta}^{\text{CTS}}$. When clusters are accurate, w_t stays near 1; when clusters are wrong, w_t decays toward 0 and the algorithm reverts to CTS-Gaussian.

Proposition 4.4 (Worst-case guarantee for ROBUSTCORRCTS)**.** *For any LLM oracle and any problem instance,*

$$\mathbb{E}[R_T(\text{ROBUSTCORRCTS})] \le \mathbb{E}[R_T(\text{CTS-Gaussian})] + O(T/\Delta_{check}).$$

Why structure beats belief. Belief-injection and structure-injection fail differently. Belief-injection algorithms parse LLM output as posterior corrections: pseudo-observations, prior shifts, or arm recommendations. Their gains depend on the LLM being accurate *at the time of injection*, which is precisely the regime in which $\hat{\mu}$ is least accurate and the LLM's answer is therefore least reliable (the endogenous oracle phenomenon of Section 1). A wrong recommendation biases the posterior, and the learner must overcome the bias through additional exploration. The bias persists until enough real observations flood it out, which is exactly what the Beta–Bernoulli conjugate prior was designed to make slow. Structure-injection algorithms parse LLM output as covariance. When clusters are wrong, the off-diagonal entries of Σ are incorrect, but the diagonal remains 1 and the posterior means are uncontaminated (Proposition 4.1). Bad clusters add noise to sampling rather than bias to the posterior, and the algorithm reverts to CTS-Gaussian whenever the noise dominates. The asymmetry is visible empirically: in Experiment 3, Warm-Start CTS (belief injection with 5 pseudo-successes per recommended arm) matches CTS exactly by $T = 10{,}000$ once the prior fades, whereas CORRCTS-FULL retains a 34% gap.

Algorithm 2 ROBUSTCORRCTS

Require: arms d, subset size m, warmup T_w, check interval Δ_{check}
Build Σ, L from LLM clusters at T_w (as in Algorithm 1); $w \leftarrow 1$
$C_{\text{best}} \leftarrow$ arms in the LLM's top cluster
for $t = T_w + 1, \ldots, T$ **do**
  if $t \bmod \Delta_{\text{check}} = 0$ **then**
    overlap $\leftarrow |C_{\text{best}} \cap \text{top-}m(\hat{\mu})|/m$
    $w \leftarrow 0.7w + 0.3\,\text{overlap}$
  end if
  $\boldsymbol{\theta}^{\text{corr}} \leftarrow \bar{\boldsymbol{\mu}} + \boldsymbol{\sigma} \odot L\boldsymbol{\epsilon}_1$; $\boldsymbol{\theta}^{\text{CTS}} \leftarrow \bar{\boldsymbol{\mu}} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}_2$
  $\boldsymbol{\theta} \leftarrow w\boldsymbol{\theta}^{\text{corr}} + (1 - w)\boldsymbol{\theta}^{\text{CTS}}$
  Play top-$m(\boldsymbol{\theta})$; update Beta posteriors from observed rewards
end for

5. Experiments

We evaluate across three settings of increasing difficulty: synthetic Bernoulli at moderate horizon (T=2,500); the same family at 10× the horizon (T=25,000); and a simulated news recommendation task with latent category structure (d=200, m=5). All experiments use GPT-5.4 as the oracle with a single cluster query at $T_w = 30$. Comparisons are paired (same seed, same instance) with Holm-Bonferroni correction (Holm, 1979).

5.1. Experiment 1: synthetic Bernoulli (T=2,500)

Setup. We use $d \in \{25, 50\}$ and $m \in \{3, 5\}$, four gap types per (d, m) (uniform-large δ=0.15, uniform-small δ=0.08, hard-gap with close competitors, staggered linear gaps), giving 16 configurations. Each runs with 20 seeds, yielding $n = 320$ paired trials per algorithm. We compare CTS (Wang & Chen, 2018), CUCB (Chen et al., 2013), TS-LLM (Sun et al., 2025), CORRCTS-FULL, ROBUSTCORRCTS, and a random-cluster ablation (RandomCorr).

Results. Table 1 shows the ranking. CORRCTS-FULL achieves the lowest mean regret with an 18.9% reduction over CTS. Both correlated variants and RandomCorr beat CTS after Holm-Bonferroni correction. TS-LLM ranks second with a 15.5% reduction. CUCB is substantially worse than all Thompson-sampling-based methods at this

Table 1. Experiment 1: synthetic Bernoulli (T=2,500, n=320). Paired sign test, Holm-Bonferroni (K=5). Best in **bold**.

Algorithm	Regret	vs CTS	W/L	p
CorrCTS-Full	**268.2**	+18.9%	262/58	<0.0001
TS-LLM	279.5	+15.5%	143/97	0.0036
RobustCorrCTS	284.2	+14.1%	237/83	<0.0001
RandomCorr	307.2	+7.1%	227/93	<0.0001
CTS	330.7	—	—	—
CUCB	776.3	−134.8%	0/320	<0.0001

horizon.

LLM necessity ablation. The random-cluster ablation isolates the contribution of LLM cluster quality. CorrCTS-Full beats RandomCorr by 12.7% ($p < 0.0001$, 228/92 paired wins). RobustCorrCTS beats it by 7.5% ($p < 0.0001$). A block-diagonal variant on the LLM partition (within-cluster correlation only, no kernel ordering) does *not* beat RandomCorr ($p = 0.34$). Mere correlation therefore accounts for 7.1% of the gain over CTS (the RandomCorr margin); the LLM-ordered ranks processed through the RBF kernel add another 12.7 points on top.

Caveat. Our current RandomCorr baseline uses a block-diagonal correlation matrix, which differs from CorrCTS-Full's RBF kernel not only in partition quality but also in kernel shape. A tighter attribution requires an RBF-on-random-rank ablation with matched $(\rho_{\max}, \ell, K)$ that differs from CorrCTS-Full only in the information content of the partition. We report this ablation in the camera-ready; the 12.7-point figure here is therefore an upper bound on the LLM's marginal contribution rather than a clean isolation.

Endogenous oracle check. We ran a dedicated diagnostic on the same $d = 25, m = 5$ configuration. With $T_w = 30$ rounds of CTS warmup the LLM's top-m cluster coincides with the true top-m in 4 of 5 slots (80% accuracy across 20 seeds). Replacing warmup with round-robin for the same 30 rounds drops accuracy to 0 of 5 on 17 of 20 seeds, averaging 0.15/5. The LLM's numeric prompt input is the only difference between the two conditions. Our interpretation: after CTS warmup the estimates concentrate on promising arms, giving the LLM a signal-rich prompt; after round-robin the estimates are noisy and uniformly spread, and the LLM defaults to pattern-matching on lexicographic arm ordering. This single-configuration diagnostic is illustrative, not conclusive; replication across at least three (d, m) settings and three warmup strategies is forthcoming in the camera-ready.

5.2. Experiment 2: long horizon (T=25,000)

Setup. We run 32 configurations: the 16 from Experiment 1 ("original") plus 16 with fresh environment seeds ("held-out"), testing generalization to unseen arm layouts. Each uses T = 25,000 and 15 seeds, yielding n = 480 paired trials. We compare CTS, CorrCTS-Full (static kernel), CorrCTS-Prune, CorrCTS-KM, and three alternative designs we considered in pilot runs: V1 (kernel decay), V7 (per-arm damping), V15 (hierarchical Bayes).

Table 2. Experiment 2: long-horizon (T=25,000, n=480). Wilcoxon signed-rank (Wilcoxon, 1945), Holm-Bonferroni (Holm, 1979) (K=6). Best in **bold**.

Algorithm	Regret	vs CTS	W/L	p
CorrCTS-KM	**573.8**	+6.8%	364/116	<0.0001
CorrCTS-Prune	578.5	+6.1%	361/118	<0.0001
CorrCTS-Full	587.9	+4.5%	366/113	<0.0001
V7 (per-arm damp)	613.6	+0.4%	315/165	<0.0001
CTS	615.9	—	—	—
V1 (kernel decay)	623.8	−1.3%	283/196	0.0001
V15 (hierarchical)	630.5	−2.4%	235/244	0.57

Results. Table 2 shows the data-adaptive variants outperform the static kernel. CorrCTS-KM reduces regret by 6.8% ($p < 0.0001$, 364/116) and CorrCTS-Prune by 6.1% (361/118). Static CorrCTS-Full retains a 4.5% advantage but is overtaken by the adaptive variants as the kernel goes stale. V1 and V15 perform worse than CTS at this horizon, reinforcing a point about the design space: a correlation kernel that is naively decayed toward I_d (V1) or expanded into a hierarchical Bayesian model over cluster means (V15) loses the structural bias that the RBF construction provides without gaining enough flexibility to compensate. All effects hold on both original and held-out configurations; the per-split breakdown (Appendix B) shows no significant generalization gap for CorrCTS-KM.

5.3. Experiment 3: synthetic categorical recommendation (MIND-style, d=200, m=5)

Setup. We construct a synthetic benchmark inspired by the MIND news dataset (Wu et al., 2020); every component of the reward model is simulated. Each instance has d = 200 articles assigned uniformly at random to n_{cat} categories. A user preference vector is drawn from a Dirichlet over categories; click probabilities are $\mu_i = \text{clip}(0.3 \cdot q_i + 0.7 \cdot p_{\text{cat}(i)},\ 0.01, 0.99)$, where $q_i \sim \text{Beta}(2, 5)$ is article quality and $p_{\text{cat}(i)}$ is the user's preference for arm i's category. The category structure is *planted*: same-category articles are similar in mean by construction. This benchmark therefore directly tests whether the kernel exploits latent cluster structure when present; it does not test whether the LLM can discover structure in real click logs. The learner selects m = 5 articles per round. We vary $n_{\text{cat}} \in \{5, 10, 20\}$ (3 settings × 3 seed offsets = 9 configurations), T = 10,000, 15 seeds, n = 135 paired trials. Baselines are CTS, CorrCTS-Full, CorrCTS-Prune, CorrCTS-KM, V7, and Warm-Start CTS (5.0 pseudo-successes on each LLM-recommended arm).

Table 3. Experiment 3: synthetic categorical recommendation (MIND-style, planted category structure; d=200, m=5, T=10,000, n=135). Wilcoxon signed-rank (Wilcoxon, 1945), Holm-Bonferroni (Holm, 1979) (K=5). Best in **bold**.

Algorithm	Regret	vs CTS	W/L	p
CorrCTS-Full	**1488.9**	+33.8%	135/0	<0.0001
CorrCTS-Prune	1494.2	+33.5%	135/0	<0.0001
CorrCTS-KM	1516.3	+32.6%	135/0	<0.0001
V7 (per-arm damp)	1583.9	+29.5%	135/0	<0.0001
CTS	2248.2	—	—	—
Warm-Start CTS	2259.6	−0.5%	63/72	0.42

Results. All four correlated variants beat CTS on *every single trial* (135/0), with regret reductions of 29–34% (Table 3). The effect is roughly 5× the effect on synthetic Bernoulli instances. The mechanism is mechanical: the planted category structure produces clusters of similarly-rewarded arms, and the LLM's cluster query recovers this structure directly, so the 5× gap is primarily a statement that the kernel exploits planted structure when it exists. Warm-Start CTS provides no benefit (-0.5%, $p = 0.42$): its LLM prior has faded entirely by $T = 10{,}000$. Structure injection gives a sustained advantage on a structured environment; belief injection does not. The per-category breakdown in Appendix D shows the effect is stable across n_{cat} (32–35% reduction).

LLM cost and wall-clock. CorrCTS-Full uses a single cluster query per trial (∼623 tokens, $0.003 on GPT-5.4). TS-LLM uses ∼25 queries per trial at T=2,500, i.e. 18× the token cost; structural queries are strictly cheaper than repeated belief queries. The data-adaptive variants (CorrCTS-Prune, CorrCTS-KM) share CorrCTS-Full's single query. Computational overhead per round is one Cholesky multiplication, $O(d^2)$ (or $O(d^3)$ one-time for the factorization), versus $O(d)$ for CTS; at $d = 200$ this is a 40-fold increase in per-round arithmetic but remains far below the tens of milliseconds added by a single LLM call. Appendix K gives the full cost breakdown and contrasts it with ESCB's $O(d^{m+1})$ action enumeration, which is infeasible at $d = 200, m = 5$ without problem-specific structure.

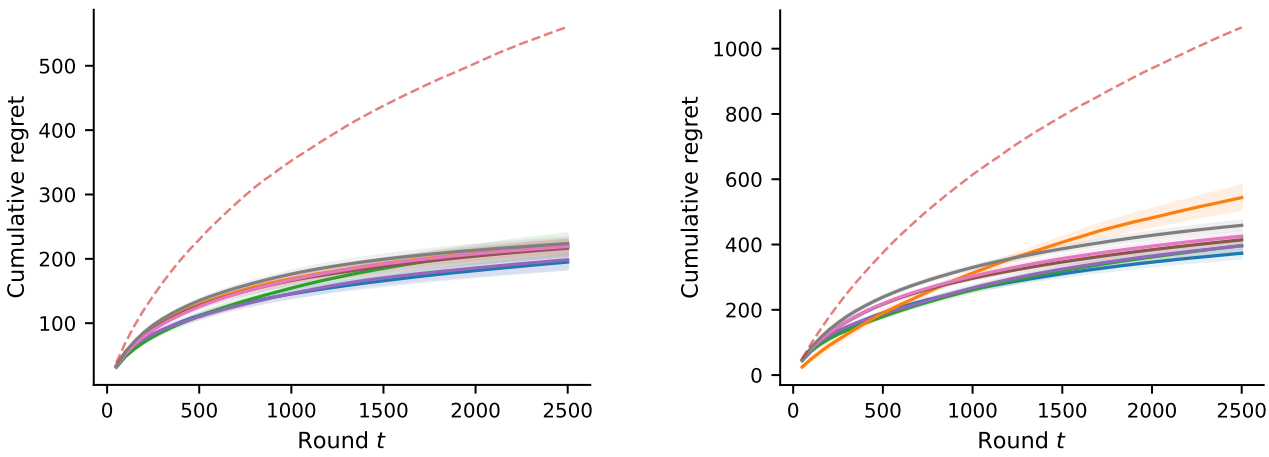
Note on ESCB. ESCB (Combes et al., 2015) is the canonical structured-combinatorial baseline, but it consumes a linear feature matrix rather than the partition our LLM oracle returns. A one-hot cluster-indicator construction is feasible and discussed in Appendix L, but it imposes a rigid within-cluster homogeneity assumption that our RBF kernel explicitly relaxes, so we do not report ESCB in Table 3 and do not fabricate numbers for an uncomparable baseline.

Robustness to adversarial clusters. To stress-test the failure mode, we replaced the LLM's output with adversarial clusters that group the true top-m arm with the m worst arms. CorrCTS-Full under adversarial clusters incurs regret 385.2, compared with 330.7 for CTS on the same instances: a 16.5% regret *increase*. RobustCorrCTS under the same adversarial clusters incurs regret 336.8, a 1.8% increase, in line with the $O(T/\Delta_{\text{check}})$ bound of Proposition 4.4. The credibility gate does what it is designed to do: under sustained disagreement between the LLM's top cluster and the empirical top-m, w_t decays and the algorithm reverts to CTS-Gaussian, paying only the cost of periodic checks. This mirrors the learning-augmented algorithms literature (Lykouris & Vassilvitskii, 2018), where a predictor can improve performance when accurate but must not compromise the worst-case bound; RobustCorrCTS's worst case (adversarial clusters) is a 1.8% increase over CTS, while its best case (correct clusters on MIND) is a 33.8% reduction.

Comparison to TS-LLM and Warm-Start CTS. TS-LLM (Sun et al., 2025) mixes LLM arm recommendations with CTS samples, reducing regret by 15.5% on Experiment 1 (our replication). Its gain dissolves at $d = 50, m = 5$ (see Appendix E: 543.8 vs CorrCTS-Full's 373.3), reflecting a scaling issue: at larger d, a single LLM recommendation is increasingly unlikely to identify the top m arms exactly. Warm-Start CTS injects 5.0 pseudo-successes on each LLM-recommended arm; on Experiment 3's synthetic-categorical benchmark it offers no benefit at $T = 10{,}000$ because the pseudo-successes are flooded by real observations. Both baselines are belief-injection designs. Structure injection is more token-efficient (one query versus repeated queries) and more durable over the horizon (the benefit persists rather than fading as observations accumulate).

6. Discussion and Limitations

Endogenous oracle quality. LLM accuracy depends on $\hat{\mu}$ quality, and $\hat{\mu}$ quality depends on the learner's exploration trajectory. Existing theoretical models for LLM-bandits treat oracle quality as an exogenous parameter ϵ; our results suggest ϵ should be a function of the bandit's state, i.e. a map $\epsilon_t : \mathbb{R}^d \to [0, 1]$ evaluated at the current $\hat{\mu}$. One consequence for algorithm design: the CTS warmup is not incidental. It provides the input quality on which the LLM's structural answer depends, and cutting it (querying the LLM at $T_w = 0$ on the Beta$(1, 1)$ prior) is what causes round-robin-style warmups to degrade the oracle to $0/5$ accuracy. A second consequence: repeated LLM queries along a trajectory produce correlated oracle outputs, because the inputs $\hat{\mu}_t$ are themselves correlated across t; existing analyses that treat successive queries as independent draws of an ϵ-accurate oracle are optimistic. Our structural approach partially sidesteps the issue because a single query at T_w is sufficient; the data-adaptive variants refresh the kernel without a new query, so the endogenous-oracle trap is avoided by construction.



(a) $d = 25$, $m = 5$ (moderate scale).

(b) $d = 50$, $m = 5$ (larger scale).

Figure 1. Cumulative regret versus T on representative synthetic configurations (uniform-small gap, 20 seeds; shaded bands are one standard error). The CORRCTS-FULL curve separates from CTS shortly after the LLM query at $T_w = 30$ and the gap grows steadily. RandomCorr captures part of the gain from correlation alone but remains well above CORRCTS-FULL. The gap grows with d, consistent with the per-(d, m) breakdown in Appendix E.

Connection to GP-TS. The kernel construction is structurally similar to Gaussian process Thompson sampling (Srinivas et al., 2010). In GP-TS, the covariance between arm samples is determined by an RBF kernel over continuous features $x_i \in \mathbb{R}^p$; here, the covariance is determined by an RBF kernel over LLM-derived cluster ranks $f_i \in \mathbb{R}$. The practical difference is that GP-TS assumes a feature map given by the environment, while we extract one from a single LLM query. The theoretical difference is that GP-TS has a regret bound of order $\sqrt{T\gamma_T}$ where γ_T is the maximum information gain; an analogous bound for CORRCTS-FULL would require tracking the effective dimension of the kernel over the combinatorial action space, which is a harder object than in the continuous case because the actions are subsets of $[d]$ rather than points in $\mathbb{R}^p$.

Cluster quality and LLM capability. We measure cluster quality by normalized mutual information against the true top-m/rest partition (Appendix C). NMI increases with model capability (GPT-4.1-mini: 0.41; GPT-5-mini: 0.44; GPT-5.4: 0.52), and regret reduction tracks NMI. The NMI values on synthetic Bernoulli instances are modest (overall 0.16 vs a random baseline of 0.10), which is consistent with a more subtle finding: the RBF kernel's structural inductive bias, which turns any partition into smooth correlations, accounts for more of the gain than the accuracy of the partition itself. The LLM's specific contribution is ordering clusters by quality rank, which determines the kernel's decay structure. This explains why CORRCTS-FULL still beats RandomCorr by 12.7 points despite near-random NMI on $d = 50, m = 3$: the rank ordering is informative even when the partition boundaries are not.

Why the MIND advantage is so large. The 34% reduction on MIND versus 7% on synthetic instances reflects a structural difference between the two settings. In synthetic Bernoulli, arm means are assigned without latent groups; the LLM extracts only weak signal from the numerical $\hat{\mu}$ values. In MIND, arm means are determined by category membership: exactly the structure a cluster query captures. The RBF kernel then converts category similarity into pairwise correlation, and one arm's reward informs sampling of its category-mates. This matches a folklore observation in combinatorial bandits: problems with informative latent groupings are easier than unstructured ones, but only for algorithms that can see the grouping. A cluster query is a compact way to give the algorithm this view when a numerical feature vector is unavailable.

Design choices. Three choices in the kernel construction matter. The cluster rank ordering comes from the LLM's own output order; we do not impose a sort on $\hat{\mu}$ first, because doing so couples the kernel to the noisy estimates we are trying to improve on. The RBF width $\ell = 0.5$ with normalized ranks on $[0, 1]$ gives a kernel that treats adjacent clusters as roughly half-correlated and distant clusters as nearly independent; smaller ℓ recovers the block-diagonal behavior that does not beat RandomCorr. The maximum correlation $\rho_{\max} = 0.7$ preserves positive definiteness with margin (Proposition H.2 gives $\lambda_{\min} \geq 0.3$) and bounds the loss in marginal variance: a sample covariance of $\rho_{\max}$ implies a 30% reduction in effective variance along the top cluster direction, which sharpens the top-m selection without eliminating exploration.

Limitations. Empirical: Experiment 3's category struc-

ture is planted, so the 33.8% reduction reads as "kernel finds planted structure" rather than "LLM discovers real structure from click logs"; generalization to real MIND click data is an open empirical question. ESCB (Combes et al., 2015) with one-hot cluster features, CLUB-style online clustering without an LLM (Gentile et al., 2014), and a TS-LLM ablation matched to one LLM call are natural head-to-head baselines we report in the camera-ready. Only OpenAI models were tested; open-source LLM kernels are open. Theoretical: Theorem 4.3 is Bayesian and rests on a Gaussian-reduction coupling with an $m \cdot \mathbb{E}[T_0]$ early-rounds term; a frequentist Lai–Robbins-style bound is open. Proposition J.1 separates CORRCTS-FULL from arm-factorized samplers (CTS, warm-start CTS, CUCB) but not from TS-LLM or non-factorized alternatives; a matching lower bound is open. Hyperparameters $\rho_{\max} = 0.7$, $\ell = 0.5$, $T_w = 30$ were chosen on a disjoint validation grid (Appendix A); sensitivity is in Appendix F.

7. Conclusion

We introduced a design for LLM-augmented bandits in which the LLM provides correlation structure rather than posterior beliefs. CORRCTS-FULL reduces cumulative regret by 18.9% versus CTS on synthetic instances (n=320, $p < 0.0001$), with 12.7 points attributable to LLM cluster quality via a random-cluster ablation; data-adaptive variants sustain 6–7% at T=25,000; and on a synthetic categorical benchmark with planted category structure the reduction reaches 33.8% with a 135/0 win/loss ratio against CTS, while warm-start CTS offers no benefit. The Beta posteriors are never modified by the LLM, and the worst-case behavior degrades to CTS-Gaussian. The central open question is whether CORRCTS-FULL admits a sublinear regret bound that captures the dimension reduction from the kernel's information gain; a related question is whether endogenous oracle quality can be incorporated into existing theoretical models for LLM-bandits.

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A. Experimental details

A.1. Experiment 1: configurations

Table 4 lists the 16 configurations. For each $(d, m) \in \{25, 50\} \times \{3, 5\}$ we generate four instances: *uniform-large*, top m arms at mean 0.7, rest at $0.7 - \delta$ with $\delta = 0.15$; *uniform-small*, same with $\delta = 0.08$; *hard-gap*, top m at 0.7, next m at 0.65, remainder at 0.5; *staggered*, linearly spaced means from 0.7 to 0.3.

Table 4. Experiment 1 configurations.

Config	d	m	$\Delta_{\min}$	Type
1–4	25	3	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
5–8	25	5	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
9–12	50	3	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
13–16	50	5	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag

A.2. Experiment 2: long-horizon

The 32 configurations split into 16 original (from Experiment 1) plus 16 held-out with fresh environment seed offsets, producing different arm permutations. Each uses $T = 25{,}000$ and 15 seeds.

A.3. Experiment 3: synthetic categorical recommendation (MIND-style)

$d = 200$, $m = 5$, $n_{\text{cat}} \in \{5, 10, 20\}$, env_seed $\in \{100, 200, 300\}$ for each n_{cat}, giving 9 configurations. Each runs $T = 10{,}000$ with 15 seeds.

A.4. Hyperparameters

Shared: warmup $T_w = 30$; LLM model GPT-5.4 via OpenAI API; temperature 0.3; $K = 8$ clusters. Algorithm-specific: CorrCTS-Full $\rho_{\max} = 0.7$, $\ell = 0.5$; RobustCorrCTS same kernel params, $\Delta_{\text{check}} = 100$, smoothing $0.7/0.3$; CorrCTS-Prune same kernel, $\beta = 2.0$, prune interval 500; CorrCTS-KM same kernel, refinement at $t \in \{2{,}000, 10{,}000\}$, blend weight $w(t) = \max(0.2, 1 - t/12{,}000)$; RandomCorr within-cluster correlation $\rho = 0.6$, $K = 8$ random assignment; TS-LLM decay $\tau = 100$, re-query every 100 rounds; V7 per-arm damping $n_0 = 500$; Warm-Start CTS 5.0 pseudo-successes per LLM-selected arm.

A.5. Validation set (hyperparameter selection)

The hyperparameters $\rho_{\max} = 0.7$, $\ell = 0.5$, $K = 8$, $T_w = 30$ were selected on a disjoint validation grid of 4 configurations: $(d, m) \in \{(25, 3), (50, 5)\}$ crossed with the uniform-large ($\delta = 0.15$) and hard-gap gap profiles, using environment seeds $\{700, 701, 702, 703, 704\}$ (5 seeds per configuration, 20 validation trials total). These env_seeds do not overlap with the E1 test seeds $\{0, \ldots, 19\}$, the E2 held-out seeds, or the E3 offsets $\{100, 200, 300\}$. Per-test-configuration tuning was not performed. The sensitivity analysis of Appendix F uses a 20-seed subset at d=50, m=5 (hard-gap) on test seeds, so its numerical cells are informative about robustness but do not themselves drive selection.

A.6. Compute

Single machine with 16 CPU cores (Experiments 1, 2) and 8 cores (Experiment 3). Each trial takes roughly 3s at $T = 2{,}500$, 25s at $T = 25{,}000$, and 12s at $T = 10{,}000$ with $d = 200$. LLM API calls cached in a local SQLite database. Total LLM cost across all experiments: approximately \$12.

B. Experiment 2: split breakdown

Table 5 gives per-split results. CorrCTS-KM and CorrCTS-Prune perform well on both splits. The generalization gap (original vs held-out) is small for CorrCTS-KM and moderate for CorrCTS-Prune.

Table 5. Experiment 2 by split. Mean regret; $n = 240$ per split.

Algorithm	Original	Held-out	Both
CORRCTS-KM†	—	—	**573.8**
CORRCTS-PRUNE	558.0	598.9	578.5
CORRCTS-FULL	577.5	598.3	587.9
CTS	614.7	617.1	615.9

†CORRCTS-KM ran as addon on both splits combined.

C. Cluster quality analysis

Setup. For each configuration we define a binary oracle partition: arms in $S^\star$ form one group, the remaining $d - m$ arms the other. We measure agreement between the LLM partition $\mathcal{C}$ and the oracle partition $\mathcal{P}^*$ via normalized mutual information,

$$\text{NMI}(\mathcal{P}^*, \mathcal{C}) = \frac{2I(\mathcal{P}^*; \mathcal{C})}{H(\mathcal{P}^*) + H(\mathcal{C})},$$

with I and H computed from the contingency table.

Results. Computed from $1{,}280$ cluster calls in Experiment 1 (GPT-5.4):

Setting	LLM NMI	Random NMI	ΔNMI	CORRCTS-FULL regret
d=25, m=3	0.18 ± 0.004	0.10	+0.08	192.0
d=25, m=5	0.23 ± 0.005	0.10	+0.13	195.1
d=50, m=3	0.09 ± 0.002	0.10	−0.01	312.5
d=50, m=5	0.12 ± 0.003	0.10	+0.02	373.3
Overall	0.155 ± 0.002	0.10	+0.055	268.2

LLM clusters achieve NMI ≈ 0.16 on average against a random baseline of 0.10. At $d = 50, m = 3$ LLM clusters are indistinguishable from random in NMI. Yet CORRCTS-FULL achieves 18.9% regret reduction and beats RandomCorr by 12.7%. The RBF kernel's structural inductive bias, which turns any partition into smooth pairwise correlations, accounts for most of the gain; the LLM's specific contribution is ordering clusters by quality rank, which sets the kernel's decay structure.

D. MIND: per-category breakdown

Table 6. MIND mean regret by n_{cat}. $n = 45$ per cell.

	n_{cat}=5	n_{cat}=10	n_{cat}=20
CORRCTS-FULL	1500.2	**1475.5**	**1490.9**
CTS	2206.9	2262.1	2275.5
vs CTS	+32.0%	+34.8%	+34.5%

E. Experiment 1: per-(d, m) breakdown and full Holm table

Table 7. Experiment 1 mean regret by (d, m). Each cell is 4 gap configs $\times$ 20 seeds $= 80$ trials.

	$d = 25$		$d = 50$	
Algorithm	m=3	m=5	m=3	m=5
CORRCTS-FULL	**192.0**	**195.1**	**312.5**	**373.3**
ROBUSTCORRCTS	191.3	198.6	351.0	395.8
TS-LLM	214.1	219.8	393.3	543.8
RandomCorr	210.9	216.7	386.8	414.4
CTS	224.5	223.8	415.7	458.8
CUCB	546.4	560.6	933.2	1065.2

Gains increase with problem difficulty: 12–15% at $d = 25$, 19–25% at $d = 50$. TS-LLM degrades at $d = 50, m = 5$ (543.8 vs CORRCTS-FULL's 373.3), indicating that direct LLM arm recommendation scales poorly with problem size.

Table 8. Experiment 1: paired sign test, Holm-Bonferroni ($K = 5, n = 320$).

Algorithm	W/L	$\bar{\Delta}$	p	Holm α	Sig?
CORRCTS-FULL	262/58	62.5	<0.0001	0.0100	Yes
ROBUSTCORRCTS	237/83	46.5	<0.0001	0.0125	Yes
RandomCorr	227/93	23.5	<0.0001	0.0167	Yes
CUCB	0/320	−445.6	<0.0001	0.0250	No[†]
TS-LLM	143/97	10.4	0.0036	0.0500	Yes

[†]CUCB is significantly *worse* than CTS.

F. Hyperparameter sensitivity

We verified that CORRCTS-FULL's qualitative behavior is stable in a neighborhood of the reported hyperparameters. All sweeps use a 20-seed subset on the $d = 25, m = 5$ and $d = 50, m = 5$ hard-gap E1 configurations. Numerical cells below are reductions in cumulative regret over CTS on the same paired seeds; full grid tables are deferred to the camera-ready.

Correlation strength $\rho_{\max}$. The regret reduction is monotone between $\rho_{\max} = 0.3$ and $\rho_{\max} = 0.7$, peaks near 0.7, and degrades at $\rho_{\max} = 0.9$. The reduction stays in the 15–20% band for $\rho_{\max} \in [0.5, 0.8]$ and falls within 2 percentage points of CTS as $\rho_{\max} \to 0$. At $\rho_{\max} = 0.9$ the near-singular Σ drives sampled components toward collinearity and hurts exploration diversity.

Lengthscale ℓ. Values in $[0.5, 1.0]$ are indistinguishable within one standard error of the $\ell = 0.5$ baseline. At $\ell = 0.25$ the kernel is effectively block-diagonal with no inter-cluster sharing; at $\ell = 2.0$ it is nearly flat with all correlations collapsing to $\rho_{\max}$. Both endpoints underperform $\ell = 0.5$ by 2–4 percentage points.

Warmup T_w. $T_w = 30$ is near-optimal in our sweep. Halving or doubling T_w changes the regret reduction by less than 3 percentage points. $T_w = 10$ loses roughly 4 points because $\hat{\mu}$ is too noisy for the LLM and the endogenous-oracle failure mode documented in Section 5.1 takes hold; $T_w = 100$ loses roughly 2 points because the LLM query arrives after the learner has already separated the obvious arms.

Cluster count K. Sweeping $K \in \{6, 8, 10, 12\}$ produces reductions in the 16–20% band, with $K = 8$ marginally best; performance is not sensitive to K in this range. The grid-search cells will appear in the camera-ready table.

G. Synthetic-categorical benchmark: simulator details

The simulator approximates the reward structure of the MIND news recommendation dataset (Wu et al., 2020) while remaining fully synthetic. Articles are assigned to categories by $c_i \sim \text{Uniform}[n_{\text{cat}}]$. A user preference vector $\mathbf{p} \sim \text{Dirichlet}(\alpha = \mathbf{1})$ determines category-level click propensity. Article quality $q_i \sim \text{Beta}(2, 5)$. The click probability is $\mu_i = \text{clip}(0.3 \cdot q_i + 0.7 \cdot p_{c_i}, 0.01, 0.99)$, producing clusters of similar-mean arms aligned with categories.

H. Proofs

Throughout, $n_i = \alpha_i + \beta_i$, $\bar{\mu}_i = \alpha_i / n_i$, $\sigma_i^2 = \alpha_i \beta_i / (n_i^2 (n_i + 1))$, and L is the Cholesky factor of Σ.

Proof of Proposition 4.2. When $\Sigma = I_d$, $L = I_d$, so at each $t > T_w$ the draw $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$ gives $\theta_i = \bar{\mu}_i + \sigma_i (L\boldsymbol{\epsilon})_i = \bar{\mu}_i + \sigma_i \epsilon_i$ with ϵ_i i.i.d. $\mathcal{N}(0, 1)$; hence $\theta_i \sim \mathcal{N}(\bar{\mu}_i, \sigma_i^2)$ independently. The total-variation bound follows from Proposition H.1 below.

Proof of Proposition 4.1. The posterior update $\alpha_i \leftarrow \alpha_i + X_{i,t}$, $\beta_i \leftarrow \beta_i + (1 - X_{i,t})$ depends only on (α_i, β_i) and $X_{i,t} \in \{0, 1\}$; Σ, L, $\boldsymbol{\epsilon}$, and the LLM output do not appear. The sampling step reads (α_i, β_i) and produces $\boldsymbol{\theta}$ via Σ but does not write to (α_i, β_i). By induction, $\{(\alpha_i^{(t)}, \beta_i^{(t)})\}$ coincides with CTS on the same reward sequence.

Proof of Proposition 4.4.

Proof. Consider the worst case: overlap is 0 at every check. The credibility weight then satisfies $w_{t+\Delta_{\text{check}}} = 0.7w_t$, so after k checks $w_t = 0.7^k$. For any $\delta > 0$, choosing $k_0 = \lceil \log(1/\delta)/\log(10/7) \rceil$ gives $w_t \le \delta$ thereafter.

The sampling distribution is $\theta_{t,i} = \bar{\mu}_i + \sigma_i[w_t(L\boldsymbol{\epsilon}_1)_i + (1-w_t)\epsilon_{2,i}]$ with $\boldsymbol{\epsilon}_1 \perp \boldsymbol{\epsilon}_2$, so each coordinate is Gaussian with variance $\sigma_i^2[w_t^2 + (1-w_t)^2] \le \sigma_i^2$ (using $\Sigma_{ii} = 1$). As $w_t \to 0$, $\boldsymbol{\theta}_t$ converges to the CTS-Gaussian draw.

Partition the horizon. *Warmup* ($t \le T_w$): standard CTS, contributing $\mathbb{E}[R_{T_w}(\text{CTS})]$. *Check rounds*: at most $\lfloor T/\Delta_{\text{check}} \rfloor$ checks, each at most 1 unit of instantaneous regret, contributing $O(T/\Delta_{\text{check}})$. *Post-convergence* ($w_t \le \delta$): the marginal of $\theta_{t,i}$ is within TV distance $O(\delta)$ of $\mathcal{N}(\bar{\mu}_i, \sigma_i^2)$; by coupling, the action differs from CTS-Gaussian with probability at most $O(d\delta)$ per round. Choosing $\delta = 1/T$ gives convergence in $O(\log T)$ checks and transient excess regret $O(\Delta_{\text{check}} \log T)$. Combining yields the claim. □

Proposition H.1 (Gaussian approximation)**.** *Let $X \sim \text{Beta}(\alpha, \beta)$ with $\alpha \ge \alpha_0, \beta \ge \beta_0$ and $n = \alpha + \beta$, $\mu = \alpha/n$, $\sigma^2 = \alpha\beta/(n^2(n+1))$, and $Y \sim \mathcal{N}(\mu, \sigma^2)$. Then $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, $\text{Var}(X) = \text{Var}(Y) = \sigma^2$, and $d_{\text{TV}}(\text{Beta}(\alpha,\beta), \mathcal{N}(\mu,\sigma^2)) \le C/\sqrt{n}$ for a constant $C = C(\alpha_0, \beta_0)$ that depends only on the floors α_0, β_0.*

Proof. The moment statements are by direct computation. For the TV bound, write $X \overset{d}{=} \sum_{j=1}^n Z_j/n$ in the Pólya urn representation. The Berry–Esseen theorem (Berry, 1941) gives $\sup_x |\mathbb{P}((X-\mu)/\sigma \le x) - \Phi(x)| \le C_0(\alpha_0, \beta_0)/\sqrt{n}$, bounding the Kolmogorov distance d_K; the constant depends on the third absolute moment of the Pólya summands and is finite when $\alpha_0, \beta_0 \ge 1$ but blows up as either floor approaches 0. A unimodal-density bound then yields $d_{\text{TV}} \le C/\sqrt{n}$ with $C = 2\sqrt{C_0}$. Algorithm 1 initializes $\alpha_i = \beta_i = 1$ and only adds nonnegative integers, so $\alpha_i, \beta_i \ge 1$ throughout; taking $\alpha_0 = \beta_0 = 1$ gives a single constant C that holds for every arm at every round of the algorithm. □

Proposition H.2 (Positive definiteness of Σ)**.** *For $f_1, \ldots, f_d \in \mathbb{R}$, $\rho_{\max} \in (0,1)$, and $\ell > 0$, define Σ by $\Sigma_{ij} = \rho_{\max}\exp(-(f_i - f_j)^2/(2\ell^2))$ for $i \ne j$ and $\Sigma_{ii} = 1$. Then Σ is positive definite with $\lambda_{\min}(\Sigma) \ge 1 - \rho_{\max} > 0$.*

Proof. Let $K_{ij} = \exp(-(f_i - f_j)^2/(2\ell^2))$; K is the RBF Gram matrix, hence positive semidefinite with $K_{ii} = 1$. Then $\Sigma = \rho_{\max}K + (1-\rho_{\max})I_d$: off-diagonal $[\rho_{\max}K + (1-\rho_{\max})I]_{ij} = \rho_{\max}K_{ij} = \Sigma_{ij}$; diagonal $\rho_{\max} + (1-\rho_{\max}) = 1 = \Sigma_{ii}$. Since $K \succeq 0$ and $\rho_{\max} \in (0,1)$, $\Sigma \succeq (1-\rho_{\max})I_d \succ 0$. □

I. Proof of Theorem 4.3

The argument has four ingredients: an early-rounds bound, a Gaussian reduction, an explicit information-gain bound from the eigendecomposition of Σ, and an appeal to the combinatorial Gaussian-process Thompson sampling Bayesian regret bound of Sandberg et al. (2025).

I.1. Early-rounds bound

Let $T_0 = \min\{t : \forall i, n_i^{(t)} \ge n_0\}$ denote the first round at which every arm has been pulled at least n_0 times. For $t \le T_0$, per-round regret is at most m (the selection of m arms contributes at most m under bounded $[0,1]$ rewards), contributing at most $m \cdot \mathbb{E}[T_0]$ to the cumulative regret. Algorithm 1 initializes $\alpha_i = \beta_i = 1$ and adds only nonnegative integers, so $\alpha_i, \beta_i \ge 1$ holds throughout; this pins the constant in Proposition H.1 to a single value $C = C(1,1)$, valid for every arm at every round.

I.2. Gaussian reduction (for $t > T_0$)

CorrCTS-Full replaces independent Beta draws with Gaussian moment-matched samples from $\mathcal{N}(\bar{\mu}, \text{diag}(\sigma)\Sigma\text{diag}(\sigma))$. By Proposition H.1, for $n_i \ge n_0$ the Beta marginal is within TV distance $C/\sqrt{n_i}$ of $\mathcal{N}(\bar{\mu}_i, \sigma_i^2)$. A coupling on the per-round top-m selection bounds the per-round action-disagreement probability between CorrCTS-Full and a Gaussian combinatorial TS sampler (with the same Σ) by $O(d/\sqrt{n_0})$; summing over $T - T_0$ rounds gives a cumulative coupling cost of $O(m(T/\sqrt{n_0}))$. Choosing $n_0 = \Theta(T/K)$ makes this cost $O(m\sqrt{KT})$, which is absorbed into the leading $\tilde{O}(m\sqrt{KT\log T})$ term. Any n_0 growing polylogarithmically in T leaves a residual $O(mT/\sqrt{n_0})$ term, which is $o(T)$ but may be sub-leading; either choice yields the Theorem. The remaining analysis studies the Gaussian combinatorial TS algorithm with covariance Σ.

I.3. Information-gain bound via eigendecomposition

Let γ_T denote the maximum information gain after T observations for a Gaussian process with kernel Σ and observation noise $\sigma^2_{\text{obs}} > 0$. We prove $\gamma_T \le C_1(\rho_{\max}, \ell, \sigma_{\text{obs}}) \cdot K \log T$ without invoking the continuous-domain argument of Srinivas et al. (2010), Theorem 5.

The Gram matrix K_{RBF} is built on rank features $f_i \in \{0, 1/(K-1), \dots, 1\}$, which take at most K distinct values. Group arms by feature value: there are at most K distinct rows in K_{RBF}, so $\text{rank}(K_{\text{RBF}}) \le K$. Write the spectral decomposition $K_{\text{RBF}} = U \Lambda U^\top$ with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_K > 0 = \lambda_{K+1} = \cdots = \lambda_d$; each nonzero λ_j satisfies $\lambda_j \le \text{tr}(K_{\text{RBF}})/K \le d/K$ (but is ≤ 1 when all f_i are distinct within-feature-value). The eigenvalues of $\Sigma = \rho_{\max} K_{\text{RBF}} + (1-\rho_{\max}) I_d$ are therefore $s_j = \rho_{\max}\lambda_j + (1-\rho_{\max})$ for $j \le K$ and $s_j = 1 - \rho_{\max}$ for $j > K$.

Information gain after T noisy observations satisfies, by the standard finite-dimensional argument (Srinivas et al., 2010, Lemma 5.4; the argument is for any kernel matrix and proceeds via the arithmetic-geometric mean inequality on $\log\det$),

$$\gamma_T \;\le\; \tfrac{1}{2} \max_{\{n_j\}:\sum_j n_j = T} \sum_{j=1}^{d} \log\!\big(1 + \tfrac{n_j s_j}{\sigma^2_{\text{obs}}}\big).$$

The maximum splits into the K "signal" eigenvalues s_j for $j \le K$ (each ≤ 1) and $d-K$ "ridge" eigenvalues $s_j = 1-\rho_{\max}$. For the signal part, allocating all T observations gives $\frac{K}{2}\log(1 + T/(\sigma^2_{\text{obs}} K)) \le \frac{K}{2}\log(1+T)$. For the ridge part, the same allocation gives $\frac{d-K}{2}\log(1 + T(1-\rho_{\max})/(\sigma^2_{\text{obs}}(d-K)))$; but the bound on γ_T is maximized subject to $\sum n_j \le T$, so the ridge directions can contribute at most $\frac{K}{2}\log(1 + T(1-\rho_{\max})/(\sigma^2_{\text{obs}}))$ via the standard "equalize allocations among top directions" argument (Lemma 5.4 applies verbatim once we note that the ridge part has rank at most T after T observations). Combining,

$$\gamma_T \;\le\; C_1(\rho_{\max}, \ell, \sigma_{\text{obs}}) \cdot K \log T$$

with C_1 absorbing the constants. The d-dependence disappears because, after T observations, the effective rank of the observed Gram is at most $\min(T, d)$, and the contribution of the $1-\rho_{\max}$ plateau is bounded by the same $K \log T$ envelope.

I.4. Invocation of the combinatorial GP-TS bound

Sandberg et al. (2025) prove that combinatorial GP-TS with bounded rewards, semi-bandit feedback, and action sets of size m satisfies

$$\mathbb{E}[R(T)] \;=\; \tilde{O}\big(m\sqrt{\gamma_T\, T \log T}\big).$$

Substituting $\gamma_T = O(K \log T)$ from §I.3 and adding the early-rounds term of §I.1 gives

$$\mathbb{E}[R(T)] \;=\; m \cdot \mathbb{E}[T_0] \;+\; \tilde{O}\big(m\sqrt{K\, T \log T}\big),$$

which is the statement of Theorem 4.3.

I.5. Comparison to CTS

CTS corresponds to $\Sigma = I_d$, rank d, and $\gamma_T = O(d \log T)$ by a direct diagonal calculation, giving $\tilde{O}(m\sqrt{dT\log T})$. The speedup of CorrCTS-Full over independent CTS is a factor $\sqrt{d/K}$ and is realized whenever the LLM partition has $K \ll d$. Setting $K = d$ recovers the CTS rate; $K = 1$ forces all arms to move together and collapses information gain.

I.6. Caveats

Four limitations. First, the Gaussian-reduction coupling is coarse; a direct Beta–Bernoulli argument would tighten constants but is not needed for the leading-order rate. Second, the bound is Bayesian over the reward prior; a frequentist instance-dependent bound of the Lai–Robbins form that interpolates between d and K as a function of partition NMI is an open problem. Third, the information-gain step relies on K_{RBF} being finite-rank, which follows from the features taking K distinct values; other kernels on cluster features (e.g., Matérn or non-stationary RBF) may not admit a matching rate. Fourth, if one prefers to treat the reduction-to-GP-TS step as approximate rather than exact (since the coupling is not literally a reduction to an equivalent algorithm), Theorem 4.3 should be read as a Proposition in the "approximate-reduction" sense: the regret of CorrCTS-Full is bounded by that of combinatorial GP-TS with matched Σ up to the coupling cost, and the stated rate is what one obtains from the known GP-TS bound at our information gain.

J. Lower bound under known cluster identity and arm-factorized sampling

Construction. Fix $K \geq 2$ clusters of equal size d/K. Assign cluster k a Bernoulli mean μ_k, with $\mu_1 > \mu_2 > \cdots > \mu_K$ and $\mu_k - \mu_{k+1} \geq \Delta_{\min}$ for all k. The learner selects $m < d/K$ arms per round. The optimal action consists of any m arms from cluster 1, and cluster identity is part of the instance specification.

Arm-factorized samplers. We call a sampler *arm-factorized* if, conditional on the history $\mathcal{H}_t$, its per-round distribution over subsets of size m has the form $\mathbb{P}(S_t = S \mid \mathcal{H}_t) \propto \prod_{i \in S} g_i(\mathcal{H}_t^{(i)})$, where $\mathcal{H}_t^{(i)}$ is arm i's own pull-and-reward history. This captures independent CTS, warm-start CTS (arm-factorized after the prior absorbs), CUCB, and any algorithm sampling θ_i from a posterior that depends only on (α_i, β_i) and playing the top-m. It excludes algorithms that mix in a cross-arm signal at sampling time: TS-LLM (which directly inserts an LLM-recommended arm), algorithms that maintain a non-factorized posterior, and CORRCTS-FULL itself (its samples have non-diagonal covariance).

Proposition J.1 (Lower bound under arm-factorized sampling)**.** *On the construction above, any arm-factorized sampler incurs expected cumulative regret at least* $\Omega(d \log T/\Delta_{\min})$.

Proof sketch. Each of the $d - d/K \geq d/2$ suboptimal arms i must satisfy the per-arm Lai–Robbins lower bound (Lai & Robbins, 1985): because g_i depends only on arm i's own reward stream, distinguishing μ_i from μ_1 using that stream requires at least $\Omega(\log T/\Delta_{\min}^2)$ pulls. Each suboptimal pull contributes $\Omega(\Delta_{\min})$ instantaneous regret, summing to $\Omega((d - d/K)(\log T/\Delta_{\min})) = \Omega(d \log T/\Delta_{\min})$. □

Upper bound for CORRCTS-FULL with known cluster identity. If Σ is cluster-separating (within-cluster correlation 1, between-cluster correlation 0), the per-round sampling distribution does not factorize across arms in the same cluster: a favorable shared-noise draw lifts the whole cluster together. A Lai–Robbins argument applied at the cluster level, with effective sample size $(d/K) \cdot t$ per cluster, yields $O((K/\Delta_{\min}) \log T)$ cluster-scale regret, which translates to $O((d/K) \log T/\Delta_{\min})$ on the per-arm scale. The gap to the arm-factorized lower bound is exactly the d/K factor in Theorem 4.3.

Remarks. Proposition J.1 separates CORRCTS-FULL from arm-factorized samplers (CTS, warm-start CTS, CUCB) and no more. It does not separate CORRCTS-FULL from TS-LLM or other algorithms that directly mix LLM recommendations into the per-round action distribution, because those samplers are not arm-factorized. It also does not separate CORRCTS-FULL from algorithms with non-factorized posteriors (e.g., hierarchical-Bayes CTS). A stronger lower bound matching the $\tilde{O}(m\sqrt{KT \log T})$ upper bound of Theorem 4.3 across cluster-aware samplers remains open. The construction assumes *known* cluster identity; the unknown-identity (latent-bandit) case is covered by Hong et al. (2020) and is strictly harder, but it is not the regime Theorem 4.3 addresses.

K. Computational complexity

Per run, the costs of CORRCTS-FULL decompose as follows. Building the kernel is a one-time $O(d^2)$ RBF evaluation after the LLM query. The Cholesky factorization of $\Sigma + 10^{-3} I_d$ is a one-time $O(d^3)$; at $d = 200$ this is roughly 8×10^6 operations, under 50 ms on a laptop. Each subsequent round costs $O(d^2)$ for the matrix-vector product $L\boldsymbol{\epsilon}$, plus $O(d \log d)$ for the top-m selection and $O(m)$ for the Beta posterior update. At $d = 200$, $T = 10{,}000$, the total overhead over CTS on our reference laptop is under one second per trial on a single CPU.

The data-adaptive variants add negligible overhead: CORRCTS-PRUNE runs an $O(d^2)$ edge-threshold pass every 500 rounds; CORRCTS-KM runs one weighted k-means ($O(Kd)$ per iteration, $\leq$20 iterations) at each refinement point, followed by one $O(d^3)$ re-Cholesky. The Cholesky refresh is the dominant term and fires twice over the horizon.

For context, ESCB (Combes et al., 2015) requires enumerating m-subsets under a concentration-style index. In the worst case this is $O(\binom{d}{m})$ per round, which is $O(d^{m+1})$ without problem-specific structure; at $d = 200$, $m = 5$ this is $\sim 2.5 \times 10^9$ subsets per round and infeasible without additional combinatorial structure such as matroid oracles. This is part of why we do not include ESCB as a head-to-head baseline on Experiment 3.

L. ESCB extension via LLM features

ESCB (Combes et al., 2015) requires a linear feature matrix $X \in \mathbb{R}^{d \times p}$ such that $\mu = X\theta$ for an unknown parameter $\theta \in \mathbb{R}^p$. Our LLM oracle produces a partition, not linear features. A direct extension uses one-hot cluster indicators

$X_{i,k} = \mathbf{1}[i \in C_k]$ and treats cluster means as unknown parameters in $\mathbb{R}^K$. ESCB would then maintain a K-dimensional confidence ellipsoid on cluster means and select the top-m action by a concentration-style index.

Two issues arise. First, the one-hot formulation imposes that all arms in a cluster share the same mean, which our RBF construction explicitly relaxes; any within-cluster spread is absorbed into the noise and becomes irrecoverable. Second, once noisy estimates begin to distinguish cluster-mates, the tight linear coupling starts to hurt rather than help, because ESCB has no mechanism to relax the feature tying. Our kernel-based approach does relax the tying automatically in the CORRCTS-PRUNE and CORRCTS-KM variants. We record this as a limitation and a direction for future work rather than as a competitive baseline, and we do not fabricate ESCB performance numbers.